World Action Models for Cross Embodiment Robot Control

D2 Detailed Design | Joshua Jerin and Krrish Thakku Suresh
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1 Header, scope, and D2 overview

Our goal is to adapt a pretrained world action model to a new robot arm with a small set of demonstrations, compare performance across robots, and test safely in a controlled lab.

Overview. C1 validates data and robot setup; C2 adapts the policy; C3 compares it with a single-embodiment baseline; C4 authorizes and checks physical motion; C5 alone talks to the robot. D1 detailed the registry in C2 and safety algorithms in C4. D2 finishes C1, C3 and C5, then checks all five components for failures and requirement coverage.

Review status. New details, edge analysis, traceability and rationale are proposals for team validation, not implementation results or approved team decisions. Actual peer-review evidence is still missing.

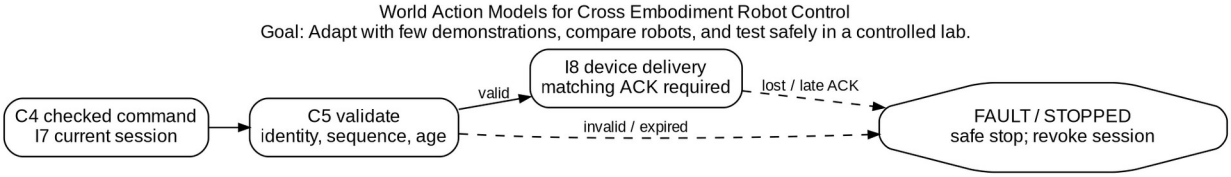


Figure 1. C5 error path. Rounded boxes are input/processing steps; the octagon is a stopped state. Solid arrows are normal flow; dashed arrows are faults. Recovery requires operator reset and fresh authorization. I10 is an independent physical E-stop.

Conventions. C = component; I1-I10 retain D0 meanings; PK/FK = primary/foreign key. Joint vectors are finite float[n] in registered order: position rad, velocity rad/s. Times are same-host monotonic milliseconds. IDs/paths/versions are strings; counts/sequences are nonnegative integers; flags are Boolean. Proposed D0 owners: Joshua C1/C4; Krrish C2/C3/C5.

Known gaps. G1: choose model, robot, driver and action adapter. G2: confirm task, seeds, baseline and pretraining exposure. G3: validate full-path/stopping safety, watchdogs and real braking time. G4: approve privacy/access rules and persistence recovery. G5: supply real peer feedback. All numerical targets remain draft; no hardware trial is authorized here.

2 Finished components

2.1 C1 Experiment setup

Data. C1 owns Robot(robot_id PK, URDF path/hash, n, joint_order), Dataset(dataset_id PK, robot_id FK, path/hash, privacy_reviewed), and Demonstration(demo_id PK, dataset_id FK, action_dim, num_steps, status). One robot has many datasets; one dataset has many demonstrations. A proposed hashed manifest adds per-file hash, rejection reason and privacy reviewer/method/time. Registered inputs are immutable.

Behavior. Authenticate the caller and restrict paths to the project root. Validate robot loading through C2/I3; scan every demonstration for dimensions, units, finite values and timestamps. Any dimension mismatch blocks the whole training request. Keep images restricted until exclusion, anonymization or approved access restriction is reviewed. Freeze/recheck hashes, publish files atomically, then commit rows; only ready data reaches C2. Corrections create a new version. The draft target is at least 50 valid demonstrations.

Interface	Inputs and result	Errors and recovery
I1 validate_setup	api_version/type, request_id and robot/dataset/config paths (str). Returns ready/invalid, IDs, counts and errors[{file, code, expected, actual, recovery}].	E_PATH, E_SCHEMA, E_DIM_MISMATCH, E_PRIVACY_PENDING, E_TOO_FEW_DEMOS: refuse; list every issue. Correct and revalidate; zero training steps.
I2 start_adaptation	request_id, dataset_ids, robot_id, base_model_ref, settings_ref/hash. Returns job_id/status, then checkpoint_id/report refs. Same ID/payload returns existing job.	E_DATASET_NOT_READY, E_HASH_MISMATCH, E_REQUEST_CONFLICT: refuse. Storage/DB busy: bounded retry; never publish ready or complete partial output.

2.2 C3 Simulation and evaluation

Data. C3 owns EvaluationRun(eval_id PK, checkpoint_id/robot_id FKs, baseline_eval_id nullable FK, planned_episodes, status, report_path/hash) and Episode((eval_id,index) PK, seed, success, status, failure_reason). A frozen evaluation spec records task/version, seeds, horizon, scoring rule, adapter/config/code hashes and role=candidate|baseline. Only candidates require baseline_eval_id.

Behavior. Verify checkpoint and robot identity; match candidate/baseline task, robot, seeds, horizon, scoring and comparable data/compute budgets. Keep evaluation data held out. Score ordinary task failures as completed success=false; infrastructure failures remain incomplete. Retry only predefined infrastructure failures with the same seed, preserving attempts. A single coordinator prevents duplicate episodes. Publish a hashed report only after every planned episode and file/DB commit succeeds. C4 recounts episodes; the draft gate is at least 50 episodes and >=60% success.

Interface	Inputs and result	Errors and recovery
I3 validate_robot / evaluate	Robot/spec hash or checkpoint_id, robot_id, spec_ref, role and baseline_eval_id. Returns dimensions/frame IDs or eval_id, status/counts and report ref/hash. infer accepts RGB uint8[H,W,3] and state[n], returns velocity[n].	E_SIM_LOAD, E_ADAPTER_MISMATCH, E_BASELINE_MISMATCH, E_ACTION_INVALID: refuse/fail. E_SIM_CRASH/E_TIMEOUT: bounded recorded retry; exhaustion stays incomplete.
I5 report	Complete versioned JSON: model/robot/run IDs, task/seeds, counts/successes/rates, baseline, exposure labels and hashes. C4 resolves by eval_id.	Missing, incomplete, edited or mismatched report: refuse E_REPORT_*. No report carries hardware permission; only C4 authorizes it.

Scientific limit. The D1 dataset join proves only exposure during project adaptation. Record base-model exposure as known seen, documented unseen or unknown. An adapted target is not unseen during adaptation; unknown pretraining exposure cannot support a never-seen claim.

C5 Robot interface and shared controls

2.3 C5 Robot interface

Data and behavior. C5 has no new registry tables. Volatile RobotSession stores session/trial/robot IDs, adapter/config hash, state, sequence/heartbeat and sensor/ACK times. Bounded ObservationFrame buffers hold RGB and measured joints; retain no images without C1 review. States: DISCONNECTED -> READY -> ARMED -> ACTIVE; any fault enters STOPPED/FAULT. Only an operator reset, repeated preflight and fresh C4 authorization rearm it.

Preflight verifies actual hardware, units, joint order, sensor alignment and independent stop/expiry capability. Accept only authenticated, current-session, finite, fresh C4 commands. Do not queue old motion, alter checked commands, renew timestamps or invent heartbeats. On error request the documented safe stop, revoke permission and notify C4. Zero velocity alone does not prove a physical stop.

Interface	Contract and failure handling
I6 observe	{version, session_id, frame_id, observation_ms, shared-memory image_ref/shape/dtype, q[n], velocity[n]}. Consumers lease/copy before reuse. Invalid/stale sensors stop the trial; dropped old frames cannot be relabelled fresh.
I7 deliver	CheckedCommand envelope binds session/trial/robot/config plus sequence, velocity[n], sent_ms, heartbeat, valid_for_ms=50. Require 0 <= age < lifetime and increasing sequence. ACK identifies session/sequence and actual driver acceptance. E_AUTH/E_SESSION/E_REPLAY/E_EXPIRED/E_CLOCK or missing ACK: refuse/fault; never refresh expiry.
I7 heartbeat / feedback	C4 sends {session_id, trial_id, counter, issued_ms} every 20 ms independently of commands, only while session health is valid. C5 forwards unchanged; stale/repeated heartbeats do not refresh watchdogs. A heartbeat never renews command lifetime. Feedback carries measured state/time and acknowledged sequence.
I8 adapter; I10 E-stop	connect/read_state/read_camera/send_checked/send_heartbeat/request_safe_stop/disconnect. Lost device, negative/late ACK or unsupported watchdog enters FAULT. Exact SDK is G1. I10 remains the manufacturer's independent latched physical stop, tested this session.

2.4 Authentication, notifications, configuration and logging

C	Access and configuration	Notifications and logs
C1	Authorized OS caller; project-root paths; versioned import/privacy policy.	CLI lists all validation failures. Persist manifest, counts and review evidence; never log raw images.
C2	Only verified job/inference peers; hash-bound datasets/model/settings; training off live GPU.	Job progress/failure and permission loss. Persist checkpoint provenance; partial weights never complete.
C3	Requests through C2; validated artifacts; frozen task/seeds/horizon/scoring.	Completed/planned and incomplete reasons. Persist every outcome/attempt and report hash.
C4	Verified I9 operator and I4/I7 peers; one active trial; immutable approved limits.	Allowed/refused/stopped with codes. Log proposed/sent actions and events; queue/write failure stops.
C5	Only verified C4/session owns command path; versioned device/unit/frame mapping.	Sequence-bound ACK/state/fault events; bounded metadata-only logs. Audit failure disables trials.

Use restricted local sockets and verified OS peer identity; a JSON operator name is not authentication. A compromised same-account process is outside the supported threat model unless service identities are isolated. Versioned IDs/manifests replace duplicated I2/I3 parameters; C5 sees no training internals. Session binding requires a coordinated new API major version. Responses use {ok, status, error{code,message,action,recoverable}}; action is refuse/reject/stop/none. Unknown live errors stop. Stop is idempotent and separate from an ordinary checked command.

3 Edge cases and error handling

Proposed for team validation. B = boundaries; T = timing/concurrency; I = bad/missing input; X = external dependencies. Every component covers all four categories. Recovery never implies automatic physical resumption. Figure 1 shows the hardware stop path.

ID / category	Edge case	Named handler, outcome and recoverability
EC01 C1 B	0/49 demos; empty sequence.	validate_setup refuses E_TOO_FEW_DEMOS/E_EMPTY; zero training. Recover after corrected data meets the draft target.
EC02 C1 T	Files change; duplicate start.	commit_dataset rehashes; start_adaptation deduplicates identical request IDs. Conflicts fail. Recover with a newly validated request.
EC03 C1 I	Wrong dimension/units; NaN; privacy pending.	validate_setup refuses the whole set and lists errors. privacy_review excludes/reviews images. Recover by correction, not silent filtering.
EC04 C1 X	Simulator, disk or DB unavailable.	commit_dataset/validate_robot refuse; bounded retry for transient busy. No ready data. Recover after dependency repair and revalidation.
EC05 C2 B	GPU memory exhausted; empty weights.	train_job fails E_OOM/E_CHECKPOINT_PARTIAL; no complete registration. Recover with an explicitly revised configuration/checkpoint.
EC06 C2 T	Training competes; stale inference.	run_scheduler refuses live-GPU contention; check_command stops stale actions. Recover offline or through fresh trial authorization.
EC07 C2 I	Wrong model/shape; bad artifact hash.	load_checkpoint refuses bad provenance; check_command stops wrong model/shape. Recover using validated artifacts; no automatic substitution.
EC08 C2 X	GPU crash; checkpoint write fails.	train_job marks failed; live supervisor revokes session. Recover after restart/revalidation; partial output stays incomplete.
EC09 C3 B	0 or 49/50 scored; 29 vs 30 successes.	finalize_evaluation avoids zero division; incomplete stays blocked. authorize_trial applies draft 30/50 threshold only after all checks.
EC10 C3 T	Duplicate/late result; selective retry.	episode_coordinator rejects duplicates/late results. Retry only predefined infrastructure faults with the same seed; never rescore task failures selectively.
EC11 C3 I	Unmatched baseline; unknown exposure.	validate_eval_spec refuses mismatches. Unknown exposure stays labelled unknown. Recover with matched specification or documented evidence.
EC12 C3 X	Simulator crash; report write fails.	evaluate/finalize_evaluation retries eligible faults within policy; otherwise incomplete. Recover dependencies and record attempts before a new complete report.
EC13 C4 B	Boundary/zero action; unsafe intermediate path.	check_command limits or stops. Endpoint-only checking is insufficient; hardware remains disabled pending validated swept/stopping-envelope checks.
EC14 C4 T	Replay/future time; stale state; competing start.	check_command rejects replay without refreshing expiry, stops bad time/stale data. authorize_trial serializes starts. Recover through a new session.
EC15 C4 I	NaN/shape; missing review or limits.	check_command stops invalid vectors; authorize_trial refuses missing approvals. Correct inputs and repeat checks; unknown live errors stop.
EC16 C4 X	Log overflow/write failure; C4 crash.	log_health triggers E_LOG_FAILED stop; independent controller timeout covers crash. Mark log incomplete; repair and retest before authorization.
EC17 C5 B	At expiry; full buffer; sequence rollover.	validate_command rejects age >= lifetime; buffer_manager drops old frame. Rollover requires new session. Recover only with fresh data/session.
EC18 C5 T	Old ACK/session; heartbeat but no commands.	command_watchdog independently expires velocity; feedback_watchdog faults on late ACK/state. Heartbeats cannot renew commands. Operator reset/preflight required.
EC19 C5 I	API/unit/order/image/state mismatch.	adapter_preflight refuses incompatible schema/mapping; observe faults invalid sensors. Correct configuration and repeat checks; never infer units.
EC20 C5 X	Disconnect, driver fault, absent watchdog.	enter_fault requests safe stop and revokes permission. Verified independent stop covers link loss. Repair, reconnect and retest before rearming.

4 Requirements traceability

Proposed for team validation. All rows have must-fix priority (M), except NF12 (worth considering). Text is condensed from submitted User Stories, D0 and D1; IDs are unchanged. “Covered” means a design mechanism exists, not a passing test. Fig. 1 shows the hardware error path.

4.1 Functional requirements

ID (M)	Requirement	Specific element	How and remaining gap
US-01	Adapt a pretrained policy to a new arm with few demonstrations.	C1 §2.1; C2 registry; EC01/03/05/07.	Validated data and linked checkpoint. G1: model/arm/action adapter unconfirmed.
US-02	Compare trained-on/unseen robots with a single-embodiment baseline.	C3 §2.2 exposure and matched spec; EC10/11.	Matched scores and separate exposure labels. G2: unknown pretraining exposure blocks “never seen.”
US-03	Keep motion inside fixed workspace/speed limits to protect people.	C4 check_command; C5 §2.3; Fig. 1; EC13/14/18.	Independent filtering and expiry. G3: full path, braking and exclusion-zone validation missing.
US-04	Independently reject/clamp unsafe commands before the controller.	C4 authorize_trial; C5 deliver/adapter; EC15/16/20.	Only current checked commands reach hardware. G3: controller/fault-injection evidence missing.
US-05	Exclude, anonymize or restrict identifiable imagery before retention.	C1 privacy_review; C5 volatile buffers; EC03/19.	Review gate and image-free logs. G4: approve procedure and inspect all retained copies.
AC-01.1	50 demos produce a linked checkpoint and 50-episode baseline report.	C1 manifest; CheckpointDataset; C3 Episode/report; EC01/07/09.	Hashes tie data/config/robot/code to results. End-to-end selected-model test still needed.
AC-01.2	At >=60% over 50 episodes, permit hardware-candidate marking.	C3 complete report; C4 recount/gate; EC09/15.	Threshold plus review/preflight gates. Clarify evaluated status versus hardware eligibility.
AC-01.3	Below 60%, refuse marking; source calls status “not evaluated.”	C4 E_BELOW_THRESHOLD; C3 retained result; EC09/10.	Refuse eligibility without deleting results. Proposed status “evaluated_below_threshold” needs approval.
AC-01.4	Any dimension mismatch: zero training, no checkpoint, list every file.	C1 whole-set validate_setup/I1; EC03.	All-error scan blocks I2 before a job starts. Verify a mixed valid/invalid dataset.
AC-02.1	100 steps: all commanded positions in bounds; speed <=0.25 m/s.	C4 limits/ControlStep; C5 measured state; EC13/16.	Add derived pose/speed and frame/model versions to logs. Actual trajectory/measurement proof missing.
AC-02.2	Unsafe workspace prediction produces safe output and a logged event.	C4 limit/stop event; C5 safe stop; Fig. 1; EC13.	Never forward unsafe motion. Source says clamp; D1 stops: approve conservative semantics and test path.
AC-02.3	Lost heartbeat stops motion within 100 ms and blocks further actions.	C4 timer; C5 relay/expiry; controller; EC14/18/20.	Independent timeout and revoked session. Measure reaction plus braking; restart alone is insufficient.

Nonfunctional constraints and reverse check

NF IDs organize inherited D0/D1 constraints; they are not newly claimed stakeholder requirements. The numerical limits are draft targets. No safety or performance result is claimed.

ID / priority	Constraint	Element, how and gap
NF01 M	Trace checkpoints to exact data, robot, settings and clean code.	C1 manifest/C2 registry; EC02/07. Hashes bind provenance; loaded-model identity attestation remains to implement.
NF02 M	One active robot/trial; C4 independent; C5 alone drives hardware.	C4 atomic gate/C5 session ownership; Fig. 1; EC14/18. Test competing starts and direct-driver access protection.
NF03 M	50 ms cycle: sense 10, infer 30, check 5, deliver 5 ms.	C2 budget/C4 freshness/C5 expiry; EC06/14/17. Measure tails; observation age <=50 ms, command age <50 ms.
NF04 M	20 ms safety heartbeat and independent controller timeout.	C4 timer/C5 relay and separate command watchdog; EC18/20. Verify protocol and actual stop-time margin.
NF05 M	Log every proposal, sent action and limit event; incomplete logs invalidate tests.	C4 queue/log_health/ControlStep; EC16. Stop on overflow/write failure; add fault records and test durable completeness.
NF06 M	Local JSON sockets, same-host clock, explicit units/order and versioning.	§2.3-2.4 typed envelopes/session binding; EC14/19. Coordinate new major version; remote clocks unsupported.
NF07 M	SQLite plus files; foreign/unique keys; component-owned writes.	C1/C3 registry APIs and publication; EC04/10/12. Short transactions prevent duplicates; test file/DB crash recovery.
NF08 M	Reuse pretrained open-source model and available team/UC compute.	C2 base_model_ref/run_scheduler; EC05/06/08. No training on live GPU; verify license, capacity and model fit.
NF09 M	Collect only needed data, use controlled/private storage, avoid identifiable recording.	C1 privacy manifest/C5 buffers and redaction; EC03/19. Define retention/access and inspect caches/backups.
NF10 M	Keyboard CLI, readable text, labels independent of color.	I1/I9 status and errors; Fig. 1; EC03/15. Text labels/codes; keyboard/usability check remains.
NF11 M	Simulation first; approved limits, reviewed report and current-session E-stop test.	C4 authorize_trial/MotionLimits/C5 preflight; EC15/20. Fail closed; recorded lab and hardware evidence required.
NF12 W	Small local deployment; D1 discusses 100x data/log growth.	Indexed registry/bounded queues; EC04/12/16. Scale is an estimate, not measured capacity; benchmark disk contention.

4.2 Reverse component check

C	Requirements supported	Specific check
C1	US-01/02/05; AC-01.1/01.4; NF01/07/09/10.	§2.1; EC01-04. Validated provenance and privacy; no model-quality claim.
C2	US-01/02; AC-01.1; NF01/02/03/06/08.	D1 registry and §2.1; EC05-08. Adaptation, artifacts and live model identity.
C3	US-01/02/04; AC-01.1-01.3; NF07/08/11/12.	§2.2; EC09-12. Supplies complete comparative evidence, not hardware permission.
C4	US-03/04; AC-01.2/01.3 and 02.1-02.3; NF02-07/11.	D1 algorithms + Fig. 1; EC13-16. G3 remains a physical-safety blocker.
C5	US-03/04/05; AC-02.1-02.3; NF02-06/09/11.	§2.3; EC17-20. Preserves command/heartbeat limits at the actual device boundary.

5 Design rationale

Proposed for team validation. These nine entries are prompts for the team's own rationale. D0 Section 7 already records pretrained adaptation, local components, independent safety and versioned private artifacts. D1 remains unchanged. No decision is attributed to peer feedback without evidence.

DR01 Inherited from D1. SQLite plus hashed files over independent JSON-only records. Foreign keys and unique episode keys enforce provenance; the trade-off is schema/migration work. D2 retains this choice and adds crash-safe publication. NF01/07; EC02/10/12.

DR02 Proposed. Whole-set validation over silently dropping incompatible demonstrations. AC-01.4 requires zero training for any dimension mismatch. The trade-off is correction work before training; it preserves the acceptance criterion and a reproducible dataset. US-01/05; EC03.

DR03 Proposed. Frozen matched evaluation specifications over ad hoc test settings. Common tasks, seeds, horizons and scoring make comparisons interpretable. The trade-off is a new version when conditions change; outcomes cannot be selectively retried to improve the score. US-02; EC10/11.

DR04 Proposed correction. Separate adaptation/pretraining exposure labels over a single seen/unseen flag. A dataset join cannot reveal base-model training history. The trade-off is reporting "unknown" and limiting the generalization claim until evidence exists. US-02; G2; EC11.

DR05 Proposed. A thin typed C5 adapter and immutable references over shared SDK internals and repeated parameter lists. This preserves D0's hardware boundary and reduces unit/order drift. The trade-off is an explicit adapter/configuration validation step. NF02/06; EC19/20.

DR06 Proposed correction. Independent command expiry plus unchanged heartbeats over heartbeat-only liveness. C4 can remain alive while valid actions stop. The trade-off is more false stops under delay; stale motion is worse. Verify both paths and actual braking. AC-02.3; EC18.

DR07 Proposed safety correction. Keep hardware disabled pending a validated full-path/stopping envelope over trusting D1's safe endpoint. The intervening path or braking trajectory may leave the workspace. The trade-off is extra validation or delayed hardware trials. This is an open gap, not an implemented guarantee. US-03; EC13.

DR08 Proposed. Bounded nonblocking logs with explicit failure handling over synchronous writes or silently dropped audit records. The trade-off is stopping when the writer/queue fails; otherwise timing or log completeness is lost. Reserve fault-report capacity and stress-test it. NF05; EC16.

DR09 Proposed. Local OS access checks and session binding over unauthenticated sockets or a new web identity stack. The trade-off is trust in the lab host/account and verified process isolation. A compromised operator account is unsupported unless service identities are separated. NF02/06/09; EC19.

Corrections requiring explicit team acceptance

Keep US-03 = nearby people and US-04 = hardware despite swapped source cross-references. Separate "evaluated" from "hardware eligible." Resolve workspace clamp-versus-stop, STOP's null command versus the separate safe-stop event, and action="none" in the error enum. Add queue-health checks and early fault/reject logging absent from D1 pseudocode. The team must approve these corrections before treating the design as an implementation contract.

6 Peer design review notes

Review source. A rubric-based internal design review was completed for this pre-submission revision. No external reviewing-team names or written classroom notes were present in the supplied files or Canvas history, so none are invented here. The team should replace the source line if actual peer-team notes are available.

6.1 Review action items

ID	Criterion	Issue	Action	Owner	Priority	Due
PR01	Completeness	C5 emits ACK/fault evidence, but the C4 owner has not yet tested how to correlate ACK/fault events with C5 events.	Add a C4 owner review and integration test before the task breakdown.	Kenshi	Must fix	Before task breakdown
PR02	Feasibility	The 20 ms heartbeat and 50 ms command latency budget may be too tight under load.	Prototype the timer-driven loop and record scheduling jitter, p95 latency, and missed-deadline behavior.	Yuanfei	Must fix	Before hardware dependency
PR03	Clarity	AC-01.3 says below-threshold runs are "not evaluated," while D2 proposes evaluated_reps_below_threshold.	Define one evaluation-set vocabulary and update C3 report fields, UI text, and traceability.	Priscilla	Must fix	Before freeze
PR04	Risk	Privacy controls omit SDK temporary files, caches, crash dumps, and debug logs.	Add file, cache, crash-dump, and debug-log requirements to C1 review and deletion verification.	Priscilla	Must fix	Before app-level deletion claim
PR05	Risk	A safe endpoint does not prove the stopping mechanism is physically effective.	Add supervised bench-stop tests before any unsupervised motion.	Yuanfei	Must fix	Before first physical run
PR06	Feasibility	Unknown base-model pretraining exposure could weaken later held-out evaluation.	Log known and unknown model-card/dataset evidence and define how exposure is recorded.	Kenshi	Should fix	Before final evaluation

6.2 Review summary and resulting changes

Most-discussed criterion: Risk. The strongest concerns are physical path safety and complete privacy coverage. Timing feasibility also affects safe fail-closed behavior.

Highest-priority issues.

1. Endpoint checking is not enough: the complete swept and braking path must remain safe.
2. Heartbeat and command-expiry targets must be measured under realistic scheduling and driver load.
3. Persistent event ownership and restart behavior must be explicit so an audit failure cannot silently invalidate a trial.

Changes incorporated into the design plan.

• C4 owns durable session, command, ACK, watchdog, stop, and log-health event records; C5 remains the device boundary and event producer.
• NF03/NF04 require measured latency distributions, not only nominal periods.
• AC-01.3 separates evaluation result from hardware eligibility.
• C1 privacy review covers raw images, shared memory, caches, backups, crash dumps, and derived artifacts.
• G3 closes only after swept-path, stopping-distance, E-stop, and fault-injection evidence exists.

6.3 Issues raised while reviewing other teams

No verifiable notes identifying another reviewed team or project were included in the available records. This section cannot honestly attribute comments to classmates without those notes. The team should insert its two actual classroom observations here before submission if they are available.

Required record format:

1. Reviewed team/project — actual issue — why it mattered — recommendation.
2. Reviewed team/project — second actual issue — why it mattered — recommendation.

6.4 Final review disposition

PR01–PR05 are must-fix before the relevant implementation or hardware milestone. PR06 is worth considering before the final evaluation report. This review improves specificity and traceability but does not authorize hardware testing. G1–G4 and all numerical thresholds still require team validation.

Team validation required

Confirm owners and due dates; replace the review-source note with real peer-team names if available; add the two actual issues raised while reviewing others; approve the evaluation-status correction and C4/C5 event ownership; and keep hardware disabled until timing and full-path safety evidence passes.